REVIEW OF THE UID452 REVISION: *THE DIRICHLET TENSOR OF TWO POLYNOMIAL COMONOIDS IS THE PRODUCT INTERNAL CATEGORY*

RICK

Author	Rick (grandparick20@gmail.com)
Date	2026-10-24 (Wake 255)
Recipient	Scot MacBeth (scot.macbeth20@gmail.com); cc Robin Langer
Title	Review of the uid452 revision of the Dirichlet-tensor note
WIP commit	This PDF corresponds to WIP commit `848fc16` (`grandpa-rick/work-in-progress`, `reviews/`), which holds this review's source. The hash was stamped in a follow-up commit, which adds this PDF.
Reviewed	`2026-10-22-day-convolution-topos-uniform.pdf`, content `32fe306`, stamp `255a884`, 8 pp. (email UID 452). I compared it against the pre-452 version and against my uid440 report (WIP `0237120`).
Sources	Shapiro–Spivak 2305.00167 (SS), full text re-read today for Def. 3.7, 3.13, 4.1, Lem. 4.3, 5.5, Thm. 4.5, 5.6 and Cor. 5.12. Lean file not run.

Bottom line. Of the 13 uid440 items, 12 are fixed and 1 is partly fixed (M1: one false display survives on page 1). Your question (1): **OK**. The equality is genuine, but three sentences should say why (SHOULD S-A). Your question (2): **a lemma**, not a standalone note. The revision introduces **3 further SHOULD** (S-B–S-D) and **3 NIT**, so 4 new SHOULD in all and no new MUST. One of them (S-B, Rem. 3.1) comes from a sentence of mine in uid440 that was too quick. The widened hypothesis is enough, and the walking-arrow example is still consistent with Theorem 1.1.

1. The uid440 items

item	status	evidence in the revision
M1	**partly**	Thm. 1.2 is gone. Example 4.1 is the walking-arrow witness, and the remark "monoidal, non-cartesian" is in. SS is described with cofunctors in §1 and §4. **But** the §1 display $\mathbf{Comon}(\mathbf{Poly}, \triangleleft, y) \simeq \mathbf{Cat}$ (p. 1) survives. So do the chain "containers $\simeq$ directed containers $\simeq$ polynomial comonoids $\simeq$ small categories" (p. 1) and "$\simeq \cdots \simeq \mathbf{Cat}$" (p. 8). With comonoid morphisms, the category is $\cong \mathbf{Cat}^\sharp$ (SS Cor. 5.12), and §4 of your own note shows that $\mathbf{Cat}^\sharp$ is not $\mathbf{Cat}$. Fix: write $\cong \mathbf{Cat}^\sharp$, or say "bijection on objects". Separately, containers (polynomials) are not equivalent to directed containers (comonoids). Drop the first $\simeq$.
M2	fixed	$(\star)$ is now "products commute with pullbacks". GK is cited only for $\mathbf{Poly}(\mathcal{E})$, $\triangleleft$ over an LCCC (p. 4). The sharpening narrative and the Δ_A^* gloss are gone.
M3	fixed	Retitled. Stated over any finitely complete $\mathcal{E}$ with exponentiable dom. Def. 2.2 makes no Day claim over $\mathcal{E}$. §7 and Table 1 are rewritten.
S1	fixed	NS Prop. 8.79 is cited by number, and the result is called a corollary of SS 4.5 + 5.6. The SS Thm. 4.5 eq. (16) and Ex. 4.6 locators are right, and Ex. 2.9 is gone. The Lem. 5.5 alternative was added, but see new item S-B.
S2	fixed	The transpose of cod (§2.2), ι on positions and naturality (§2.3), the iso-class remark (§1), and the citation of the *proof* of SS 5.6 (Step 3) are all in. The wording is still imprecise in places; see §2.
S3	fixed	Lucyshyn-Wright is removed. The Vec paragraph is softened, but it now contains an inaccurate sentence (new S-C).
S4	fixed	The witness is framed as an illustration, both caveats are stated, "first Lean certificate" is dropped, and the URL is given.

Date: 2026-10-24.

N1–N6	fixed	$\Sigma_{A_p}\Pi_{s_p}B_p^*X$ is fixed. C_{pt} is now the positive control. The [5, Prop. 5.2] citation is replaced by SS 5.6/Rem. 5.7. "Ours" is consistent. Table row 1 is fixed. Eq. (16) is not reproduced.

2. Question (1): is Step 3's "on the nose" a genuine equality? Verdict: OK (with a SHOULD on wording)

What the objects and morphisms are. SS Def. 3.13 is explicit. The *objects* of $\mathbf{Poly}_{\mathcal{E}}$ are polynomials, i.e. actual exponentiable maps $s\colon P^* \to P$ of $\mathcal{E}$. They are not isomorphism classes. *Morphisms* are isomorphism classes of diagrams (SS Def. 3.7), "where the relevant isomorphisms are between choices of the pullback $P \times_Q Q^*$". An object of $\mathbf{Comon}(\mathbf{Poly}_{\mathcal{E}})$ is a triple (c, ε, δ). So $c \otimes d = (C \times D)^\sharp$ in **Comon** means three things: (a) the two exponentiable maps are equal, (b) the two counits are equal iso-classes, and (c) the two comultiplications are equal iso-classes.

(a) holds literally, given one convention. $c \otimes d$ is $\mathrm{dom}_C \times \mathrm{dom}_D$, formed with the chosen binary products of $\mathcal{E}$ (SS Def. 4.1). Suppose $C \times D$ means the *componentwise* internal category, $(C \times D)_i := C_i \times D_i$, formed with the *same* chosen products. Then its source map is literally the same arrow of $\mathcal{E}$. Theorem 1.1 does specify the components, so this is fine. But "the product $C \times D$ in $\mathbf{Cat}(\mathcal{E})$" names an object only up to unique isomorphism. For any other product, the statement is an isomorphism, not an equality. Say once: "all limits are the chosen ones, and $C \times D$ is the componentwise product formed with them".

(c) holds as an equality of iso-classes. Both $\delta_{c\otimes d} = \iota \circ (\delta_c \otimes \delta_d)$ and $\delta_{(C\times D)^\sharp}$ are morphisms $c \otimes d \to (c \otimes d) \triangleleft (c \otimes d)$, between literally the same two objects, by (a).

- *Positions.* Both forward maps land in the one chosen object $\Sigma\Pi_{\mathrm{dom}}(\cdots)$. Each is the transpose of a map out of $C_1 \times D_1$. Transposes are unique, so the two are equal as arrows of $\mathcal{E}$ exactly when $\mathrm{cod}_{c\otimes d} = \mathrm{cod}_C \times \mathrm{cod}_D$. That is your naturality line. No isomorphism is involved.
- *Directions.* The backward leg of ι is the canonical comparison $\kappa\colon (C_1 \times D_1) \times_{C_0\times D_0} (C_1 \times D_1) \to (C_1 \times_{C_0} C_1) \times (D_1 \times_{D_0} D_1)$, modulo the choice of pullback. That choice is exactly what Def. 3.7 quotients by. The backward leg of $\delta_c \otimes \delta_d$ is $\mathrm{comp}_C \times \mathrm{comp}_D$. So the backward leg of $\delta_{c\otimes d}$ is $(\mathrm{comp}_C \times \mathrm{comp}_D) \circ \kappa$. And the composition of the componentwise product category is *defined* on the chosen pullback of products as $(\mathrm{comp}_C \times \mathrm{comp}_D) \circ \kappa$. There is nothing else it could be. So the two backward legs are the same arrow, not isomorphic arrows.

The SS dictionary (proof of Thm. 5.6) reads $\delta^\sharp$ through an isomorphism $\coprod\coprod c[\delta_1(x)(y)] \cong C^* {}_t\times_c C^*$. That too is a change of pullback representative, absorbed by Def. 3.7.

(b) holds, but the one visible isomorphism is in your definition of ε. $\varepsilon_c \otimes \varepsilon_d$ lands in $y \otimes y = (1 \times 1 \to 1 \times 1)$. This equals $y = (1 \to 1)$ only if the chosen 1×1 is 1. Your §3 writes "into $y \otimes y \cong y$". So $\varepsilon_{c\otimes d}$ is $\lambda \circ (\varepsilon_c \otimes \varepsilon_d)$, with λ the unitor. This is the correct definition of the tensor of comonoids in a normal duoidal category. Its iso-class has forward leg ! and backward leg $\mathrm{id}_C \times \mathrm{id}_D$, which is literally $\varepsilon_{(C\times D)^\sharp}$. So (b) is an equality too.

Verdict on (1): OK. The equality is genuine. No coherence isomorphism is suppressed, because the only two candidates, κ and λ, are respectively part of the *definition* of $\mathrm{comp}_{C\times D}$ and part of the *definition* of $\varepsilon_{c\otimes d}$. But the text does not yet say this:

S-A. (SHOULD) Three wording fixes. (i) In §1, "an equality of polynomials is a genuine equality of such classes" conflates objects with morphisms. Polynomials are objects, and equality of objects is literal (given chosen products). Only equality of the structure *morphisms* is an equality of iso-classes. (ii) In Step 3 (Composition), "$\mathrm{comp}_{c\otimes d} = \mathrm{comp}_C \times \mathrm{comp}_D$ under the canonical isomorphism $(\star)$" reads exactly like a suppressed coherence iso, which is what you asked me to check. Replace it with: "κ is the backward leg of ι. By definition of the componentwise product, $\mathrm{comp}_{C\times D} = (\mathrm{comp}_C \times \mathrm{comp}_D) \circ \kappa$. Hence the backward legs are equal." (iii) Add the chosen-limits convention of (a). Mention λ in the definition of $\varepsilon_{c\otimes d}$ (or choose $1 \times 1 = 1$).

A small simplification follows from (a)–(c). Once the triple equals $(C\times D)^\sharp$, the comonoid laws are immediate: $(C \times D)^\sharp$ is a comonoid by the forward direction of SS Thm. 5.6, applied to the internal category $C \times D$. The appeal to the one-by-one matching inside SS's proof is now unnecessary. (I asked for it in uid440 when Rem. 3.1 argued differently.)

3. Question (2): standalone note or lemma? Verdict: a lemma

Blunt version. The Set case is NS Prop. 8.79, word for word. The internal case unwinds SS Def. 4.1, Thm. 4.5 and Thm. 5.6, and its proof is half a page (Steps 1–3, which §2 above compresses further). Example 4.1 makes concrete a sentence that NS already print. Table 1 and §7 are accounting. None of this is wrong now, and the accounting is honest. But 8 pages of framing around a half-page corollary will read to a referee as a remark. A referee will also note that SS's Lem. 5.5 technique makes the general-$\mathcal{E}$ step routine. The rhetoric in §1 ("a question that inverted into its own answer") and §7 ("what this buys the chain") promises more than the result delivers.

Recommendation: state it as a lemma, with Example 4.1 as the following remark, in whatever unit uses internal polynomial comonoids in presheaf or sheaf models (your Graph models are the natural host). It would become a note-grade standalone only with a second result of its own. Two candidates, both unchecked for novelty: (a) describe the actual categorical product in $\mathbf{Cat}^\sharp(\mathcal{E})$, or show it fails to exist over general $\mathcal{E}$. NS remark that $\mathbf{2} \times^\sharp \mathbf{2}$ has infinitely many objects. An internal version very likely needs infinite colimits in $\mathcal{E}$, so the base hypothesis would finally do real work. (b) Prove the Day-convolution identification over Grothendieck $\mathcal{E}$ in a stated sense (uid440 option (b)).

4. New issues introduced by the revision

S-B. (SHOULD) Rem. 3.1, the "three-line alternative", has a gap. It contradicts the note's own §1 and §6. "Pointwise, the claim is the Set case [2, Prop. 8.79], applied at each object" is unjustified. SS Lem. 5.5 transports structures to $\widehat{\mathcal{A}}$. But there $c \triangleleft c$ is *not* computed at each stage: SS Ex. 3.21 and diagram (29) use $\mathrm{Hom}_{\widehat{\mathcal{A}}}(c[x], C)$, not $\mathrm{Hom}_{\mathbf{Set}}(c_a[x], C_a)$. Evaluation at a does not preserve Π_{dom}. Your §1 ("not computed objectwise in a presheaf topos") and §6 caveat say the same. So the Set case at each a concerns a different comonoid, and the reduction proves nothing beyond "it suffices to prove Theorem 1.1 in $\widehat{\mathcal{A}}$". In $\widehat{\mathcal{A}}$ you would redo Step 3. This is my fault: uid440 S1 said "in presheaves the category side is pointwise", which is true but does not finish the argument. Delete Rem. 3.1, or cut it to that one true sentence.

S-C. (SHOULD) The §1 Vec paragraph: "Poly(Vec) has no standard construction" is not right, and the precise fact is better. **Vec** is finitely complete, so SS applies verbatim. But in **Vec** *the exponentiable maps are exactly the isomorphisms.* Proof: $s^*\colon \mathbf{Vec}/A \to \mathbf{Vec}/B$ must preserve the coproduct $A \oplus A \xrightarrow{+} A$ of id_A with itself. The comparison $B \oplus B \to (A \oplus A) \times_A B$, $(b, b') \mapsto (sb, sb', b + b')$, sends $(b, -b)$ to $(sb, -sb, 0)$, so injectivity forces s to be injective. Surjectivity needs every (a, a', b) with $a + a' = sb$ to have $a \in \mathrm{im}\, s$, and a is arbitrary, so s is surjective. Hence the internal categories with exponentiable dom are the discrete ones (2-term complexes with zero arrow part). Theorem 1.1 holds over **Vec** but is degenerate there. That is the exact location of the "Vec obstruction": it is a shortage of exponentiable maps, not a lack of a construction.

S-D. (SHOULD) Example 4.1 describes the projections as functors. "π_C, π_D are the cofunctors $(f, h) \mapsto f$ and $(f, h) \mapsto h$ on morphisms" goes the wrong way. A cofunctor acts *backward* on morphisms. $\pi_C = \lambda \circ (\mathrm{id}_c \otimes \varepsilon_d)$ lifts $f\colon x \to x'$ at (x, y) to (f, id_y), because the backward leg of ε_d is id. That lift is the one the argument actually uses two lines later, when H must lift (a, id). Rewrite it as "lifting f at (x, y) to (f, id_y)".

NITs. (N-a) The Lean URL in §6 runs off the right margin and is visibly cut ("... convolution-grap"). Use `\url` with `xurl`, or break it. (N-b) Example 4.1 uses italic E for the test category next to the calligraphic base $\mathcal{E}$. Rename it T. (N-c) Step 2 says the content of ι on directions "is" $(\star)$. Say "is the comparison map κ of $(\star)$". That matters for S-A(ii).

Consistency checks (no issue found).

- *Is "any finitely complete $\mathcal{E}$" enough?* Yes. $\mathrm{dom}_C \times \mathrm{dom}_D = (\mathrm{dom}_C \times 1) \circ (1 \times \mathrm{dom}_D)$ is a composite of pullbacks of exponentiable maps, hence exponentiable (SS Lem. 4.3). $\triangleleft$ is monoidal on $\mathbf{Poly}_{\mathcal{E}}$ for finitely complete $\mathcal{E}$ (SS Thm. 3.26). ι and normal duoidality exist there (SS Thm. 4.5). The comonoid–category bijection holds there (SS Thm. 5.6). Steps 1–3 use only chosen finite limits and the universal property of Π_{dom}, which exists by hypothesis. Nothing needs local cartesian closure, coproducts or a subobject classifier.
- *Walking arrow against Theorem 1.1.* The theorem now asserts only an identification of the comonoid $c \otimes d$, which is an object of **Comon**. It asserts no universal property. Example 4.1 refutes only universality among cofunctors. The abstract, §1, §4 and §7 all say "monoidal, non-cartesian". I re-ran the witness by hand. $H(e) = (0, 0)$ and $H(x) = H(z) = (1, 1)$ are forced. $\pi_C H = F$ forces the lift of (a, id) at e to be u. But $H(\mathrm{cod}\, u) = (1, 1) \neq (1, 0)$. Contradiction, as stated. The surviving false sentence is the §1 display in M1, not the theorem.
- *Cat($\mathcal{E}$) product.* The sketch argument in §4 is correct. The projections are internal functors.